Small Causal Graphs

46 graphs of 2 to 6 nodes. The world is the graph with every edge intact. The release is the same graph sampled with a rule's edges cut on a share of rows (the cut strength, 0 to 1); a cut parent is shuffled, as DECAF does. Every edge explains a 0.25 share of its child's latent variance, lowered so no child's parents add to more than 0.9. S and Y are binary, 50/50; every other column, U included, is continuous and normal. Hidden U is in the world but never in the release, and cuts see only the visible columns. 20,000 rows, mean of 5 seeds.

FTU **DemPar** **CF** Cut marks sit in the order FTU, DemPar, CF along each edge. FTU cuts S→Y. CF cuts the non-admissible edges at Y that are tied to S given A. DemPar is CF with no admissible columns.

Panels, x = cut strength from 0 to 1. A recipient fits logistic regression for Y on the release (every released column, S included) and is scored on a fresh table from the world.
Fairness gap: difference in mean predicted P(Y=1) between the S groups on world data, in percentage points (axis 0 to 90). S inference: Brier skill of an attacker fit on the release to guess S from the other released columns, scored on world data (0 = no better than the base rate; axis -0.1 to 0.9).
Utility: Brier skill of the recipient's model for Y on world data (axis -0.1 to 0.9). Lines use the rule colors; wider, lighter lines sit underneath, so rules that cut the same edges show as nested bands. Every line starts at the same point, strength 0.

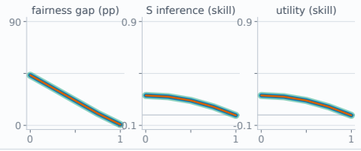
Basic shapes

Two and three nodes: the building blocks.

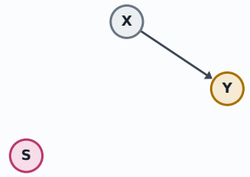
1 **Direct effect**



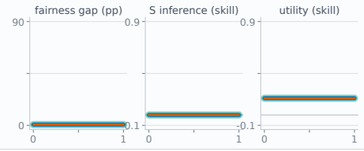
S causes Y and nothing else.



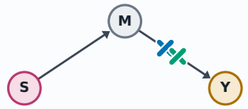
2 **Unrelated**



Y has a cause; S is not connected to anything.



3 **Mediator**



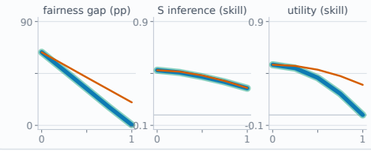
S reaches Y only through M.



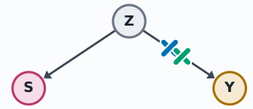
4 **Direct and mediated**



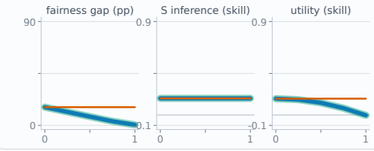
S reaches Y directly and through M.



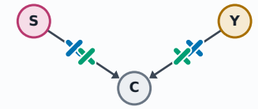
5 **Observed confounder**



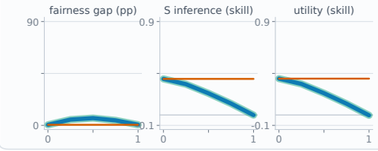
Z causes both S and Y. S has no effect on Y.



6 **Collider**



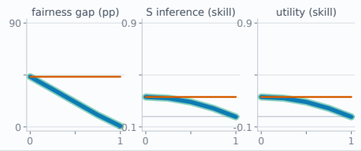
S and Y both cause C, and are independent of each other.



7 **Outcome causes S**



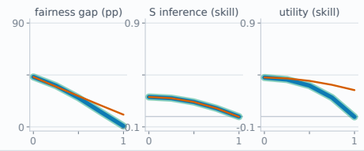
The arrow points from Y to S.



8 **Y has a child**



S causes Y, and Y causes a later column X.



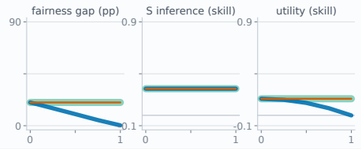
Proxies and admissible paths

A marks a column the analyst allows to carry S's influence; P marks a proxy that is not allowed.

9 Admissible mediator



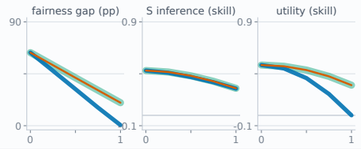
S reaches Y only through A.



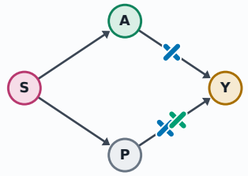
10 Admissible plus direct



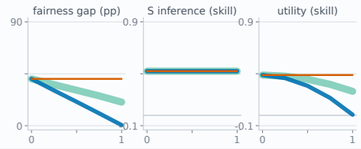
S reaches Y through A and directly.



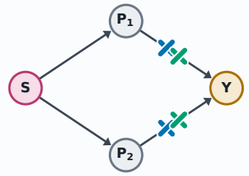
11 Proxy beside admissible



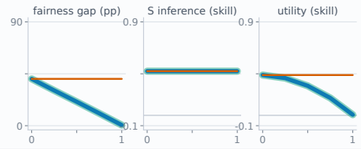
One allowed path through A and one proxy path through P.



12 Two proxies



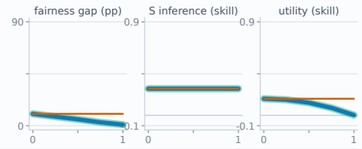
Two parallel proxy paths.



13 Proxy chain



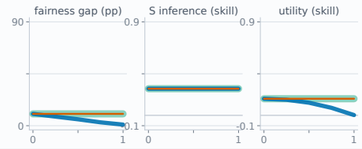
S reaches Y through two proxies in a row.



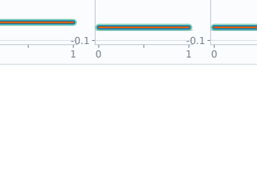
14 Proxy feeds admissible



S shapes A through P, and only A touches Y.



15 Proxy mixes S with a real cause



P is caused by S and by X; only X causes Y.

16 Admissible collider



A is caused by S and by Z, and Z also causes Y.

Hidden variables

Dashed nodes are real causes that never appear as columns.

17 Hidden confounder

U causes both S and Y; there is no visible edge between them.

18 Hidden mediator

The only path from S to Y runs through an unrecorded column.

19 Hidden cause of Y

U adds noise to Y and is unrelated to S.

20 Hidden confounder plus direct

U causes S and Y, and S also causes Y.

21 Hidden confounder of a mediator

U causes the mediator M and Y.

22 Hidden root behind a proxy

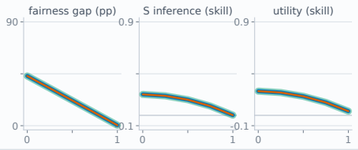
U causes S and P; P causes Y.

Hidden variables (continued)

23 Recorded label



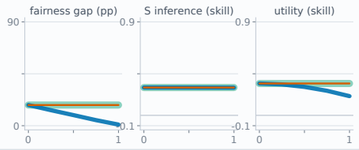
T is the true quality and is never recorded. Y is the label, and S changes it.



24 Hidden cause of an admissible column



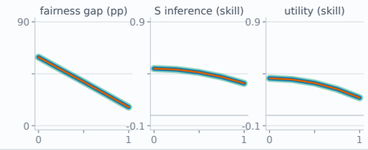
A is caused by S and by U, and U also causes Y.



25 Instrument with hidden confounder



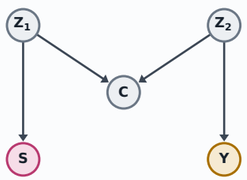
I moves S only; U confounds S and Y.



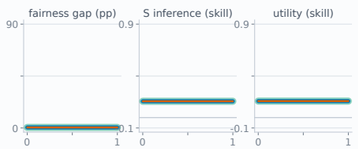
Colliders and selection

Shapes where a shared child or a conditioned column opens a path.

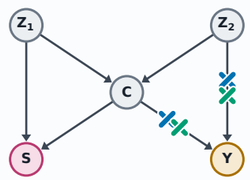
26 M-bias



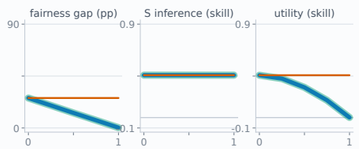
C is a child of S's cause and of Y's cause.



27 Butterfly



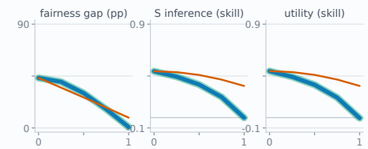
M-bias where C also causes S and Y.



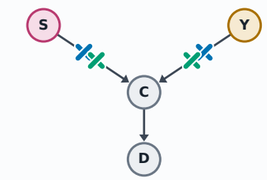
28 Shared child plus direct



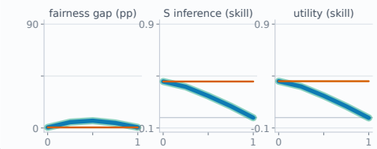
S causes Y, and both cause C.



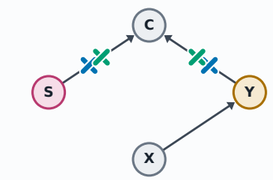
29 Descendant of a collider



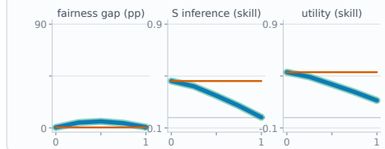
S and Y cause C, and C causes D.



30 Selection record



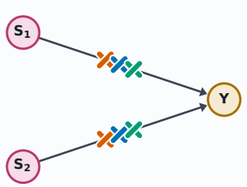
C records who was selected; it depends on S and Y. X causes Y.



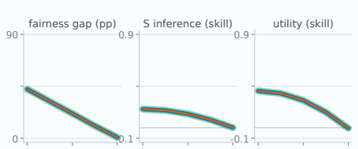
Several protected attributes

Two protected columns, as in race and sex.

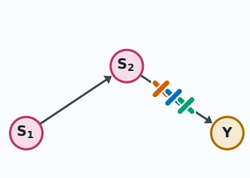
31 Two independent attributes



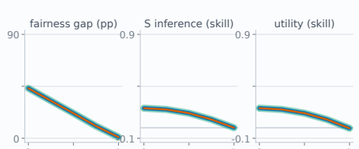
S1 and S2 each cause Y.



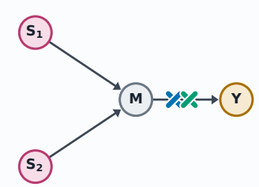
32 One attribute causes another



S1 causes S2, and S2 causes Y.



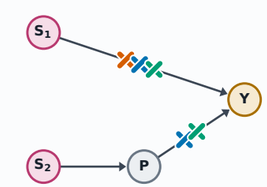
33 Intersection through a mediator



S1 and S2 meet in M, which causes Y.



34 Direct and proxy



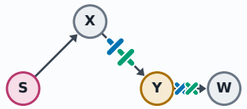
S1 causes Y directly; S2 acts through a proxy.



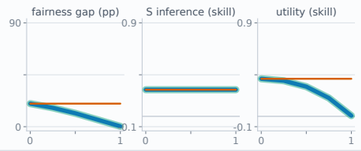
After the outcome

Columns recorded after Y, which a downstream model might still use.

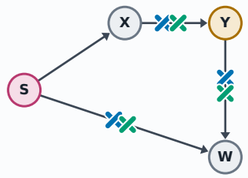
35 Later feature



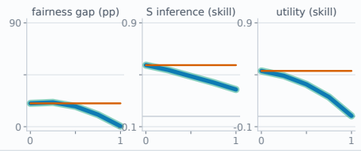
S reaches Y through X, and Y causes W.



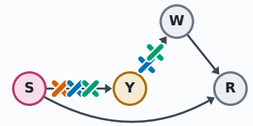
36 Later feature that also depends on S



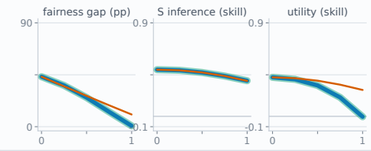
W comes after Y and also depends on S.



37 Outcome as a mediator



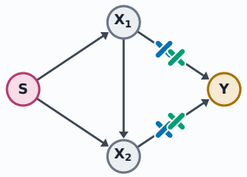
S causes Y and a score R; Y causes W, and W also feeds R.



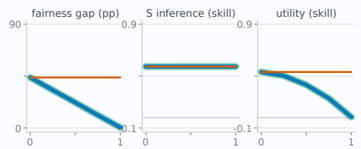
Five and six nodes

Combinations of the shapes above, including two shaped like project datasets.

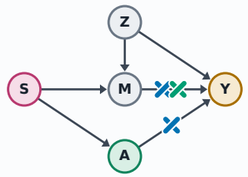
38 Diamond with a cross link



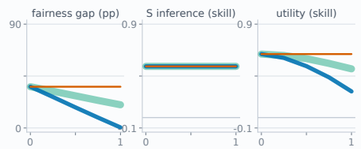
Two paths from S to Y, with X1 also causing X2.



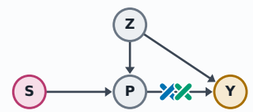
39 Census-style



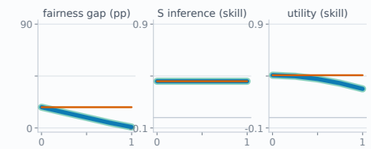
S sex, M marital status, A education, Z age.



40 Recidivism-style



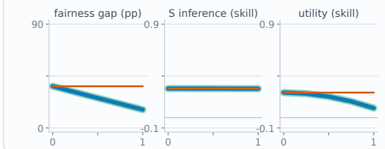
S race, Z age, P prior counts.



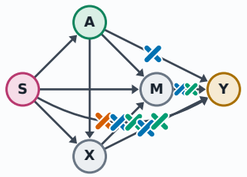
41 Front door



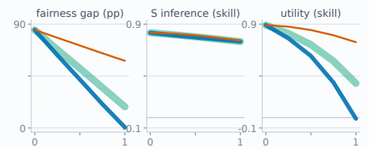
U confounds S and Y; all of S's effect passes through M.



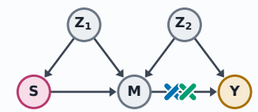
42 Every forward edge



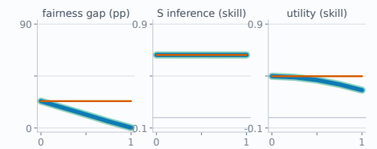
All 10 edges allowed by the order S, A, X, M, Y.



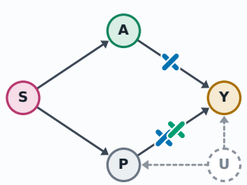
43 Confounded chain



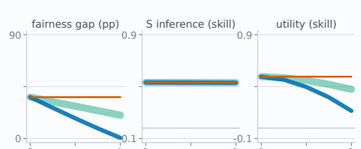
Each link of S to M to Y has its own observed confounder.



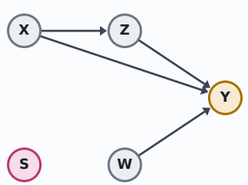
44 Proxy, admissible and hidden



U causes the proxy P and Y.



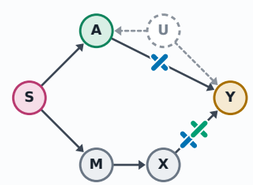
45 Isolated S in a rich world



Four columns shape Y; S touches none of them.



46 Two routes, one hidden



S reaches Y through A and through M then X; U causes A and Y.

